# RAPHAEL PELERIN

SOFTWARE ENGINEERING · SEEKING AN AI INTERNSHIP — FROM MARCH 2027

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## PROFILE

**Currently** in my fifth year of the software engineering master's at Epitech in Nice, graduating in 2027, after a year of study at Beijing Jiaotong University. **I am looking for** a six-month end-of-studies internship in **AI engineering, starting March 2027**. I want to do applied work with models, data and APIs in a real product, mostly in Python, and I would like to join a team where a full-time offer is possible at the end of the internship.

## PROJECTS

### Syntheza — Lead Frontend Web
2025 — 2026
syntheza.ovh · team of 3, three countries

- Strategic-watch platform that ingests RSS at scale, **deduplicates** what it finds and returns a daily synthesis instead of another feed.
- I own the web client: interface, design system and the React component layer. Scope was set by a **field study of 20 professionals** before any code was written.

React 19 · TypeScript · Tailwind · Radix UI · Node · Express · Prisma · PostgreSQL · React Native/Expo

### Forge — solo
2026
github.com/RaphaelPelerin/forge-asset-generator

- Game-asset generator running **SDXL and ControlNet on one local 8 GB GPU**, exposed over a FastAPI service so a pipeline can call it like any endpoint. Fully offline — no account, no image leaving the machine.
- Solved consistency with **recipes**: presets owning style, sampler and post-processing, so assets generated months apart still match.
- Measured where the model fails rather than prompting harder — the text encoder reads English only, so French subjects failed 6/6 until I added a local translation step. **~4,100 lines, 59 tests.**

PyTorch · CUDA · diffusers/SDXL · ControlNet · LoRA · FastAPI · Pydantic · pytest · React

### Keys — VST3 sampler, solo
2025
github.com/RaphaelPelerin/keys-gpt-vst

- Single-sample VST3 instrument on JUCE: ADSR envelope, three-mode filter tracking that envelope, ping-pong looping across **16 voices**.
- Every knob, curve and meter drawn with the JUCE Graphics API — **no GUI library at all**.

C++ · JUCE 8 · real-time DSP · IIR filters · Projucer

### Keys GPT — solo
2025 — 2026
github.com/RaphaelPelerin/keys-gpt

- Describe a part — mood, key, tempo, length — and get a playable piano roll and a downloadable MIDI file. Part type is **inferred from the prompt**, not picked from a menu.

React · TypeScript · OpenAI API · Web Audio · MIDI

## EXPERIENCE

### Appolonia — Front-end developer, intern
Apr — Aug 2025
Casino floor management systems

- Rebuilt the **KT Updater** module and built a group-management admin panel automating terminal shutdown and locking — software where a mistake takes a slot machine offline.
- Built an internal email-generation tool from scratch.

Angular 13 · Ionic 6 · TypeScript · SOAP/XML APIs · GitLab

### ARD — Mobile development intern
2023

- Angular application managing **NFC cards** and user profiles for alarm systems, including card scanning to display cardholder information.

### Intel · AI for Youth — competition
2022

- Computer vision and object recognition in Python with TensorFlow. **2nd place**, team category.

## EDUCATION

### Epitech Technology
2022 — 2027 · Nice
Master's, Computer Engineering. Functional and object-oriented programming, Unix, assembly, systems and network programming. Specialisations in DevOps, AI, web/mobile and cybersecurity.

### Beijing Jiaotong University
2025 — 2026 · Beijing
Fourth year abroad, software engineering, alongside Mandarin to HSK 3.

### Lycée Simone Veil
2022 · Baccalaureate
Computer Science (NSI) and Mathematics.

## CORE STACK

PyTorch · Python · C++ · React · TypeScript · Node.js · Angular · PostgreSQL · Docker · Git

## ALSO WORKED WITH

**AI & GENERATIVE**
diffusers/SDXL, ControlNet, LoRA, CUDA, OpenAI API, Anthropic Claude API, Gemini, TensorFlow

**WEB & MOBILE**
React Native, Ionic, Expo, Tailwind, Vite

**BACKEND**
Express, FastAPI, Flask, Prisma, REST, SOAP/XML

**DATA & INFRASTRUCTURE**
MongoDB, MySQL, GitLab CI

**SYSTEMS & AUDIO**
JUCE, real-time DSP, Unix, assembly, network programming

## LANGUAGES

| | |
|---|---|
| French | Native |
| English | Bilingual |
| Mandarin | HSK 3 |
| Italian | Basic |

## INTERESTS

Skiing and taekwondo · motorsport · music production in a DAW, which is where the sampler came from · travel, most recently a year in Beijing.

Full driving licence.